Problem Solutions

Date	27 JUNE 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	

Problem Statement:

Farmers often face challenges in selecting the right crop, predicting crop yield, and efficiently utilizing available resources. Traditional farming decisions are frequently based on experience rather than data, which can lead to lower productivity, higher costs, and reduced profitability. Environmental factors such as soil quality, rainfall, temperature, and humidity further complicate decision-making. There is a need for an intelligent system that analyzes agricultural data and provides accurate, data-driven recommendations.

Solution Features:

Problem	**Opti-Crop Solution**
Difficulty selecting the right crop	Recommends the most suitable crop based on soil and weather conditions.
Low agricultural productivity	Predicts crop yield using machine learning algorithms.

Problem	**Opti-Crop Solution**
Poor understanding of environmental factors	Analyzes soil, rainfall, temperature, and humidity data.
Inefficient use of water and fertilizers	Suggests optimized resource utilization strategies.
Lack of data-driven farming	Provides recommendations based on historical and real-time agricultural data.
Difficulty identifying production trends	Detects patterns and relationships in agricultural datasets.
Manual analysis takes time	Automates data analysis and prediction processes.
Limited decision support	Generates insights and recommendations for better farming decisions.